Mediant Trees for Rational Approximation and Sturmian Word Generation

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Abstract

Given an oracle for a real number θ of sufficient precision, we present four contributions. First, an algorithm that constructs a *mediant tree* of increasingly precise rational approximations to θ: starting from a Farey-adjacent bounded pair bracketing θ, each iteration applies a weighted mediant step with an optimal integer weight w^+ given by a closed-form formula (Theorem 8.1). The denominator structure mirrors the classical continued-fraction recurrence (Section 10), though a proof of superlinear growth remains open. Second, a Sturmian transduction that converts the mediant tree into a *mediant word* — a string over $\{\mathsf{O}, \mathsf{E}\}$ encoding the full approximation path — and establishes that this word is identical to the Sturmian word with slope $1/(1+\theta)$ at all interior positions, with the terminal symbol of each finite prefix fully characterised by whether r is an overestimate or underestimate of θ (Theorem 12.6). Third, we prove two structural invariants of mediant words: every non-leaf word begins with EE (Theorem 12.4), and the interior slice $\pi(r)[1{:}L{-}1]$ of any mediant word of length $L \geq 2$ is a palindrome (Theorem 12.5). Fourth, we show how to extend both algorithms to real numbers $\theta \in (0, 1)$ and $\theta < 0$ via simple algebraic reductions to the $\theta > 1$ case.

DRAFT

Contents

1 Introduction **3**

I Background **4**

2 The Stern–Brocot Tree and Mediants **4**
Definition 2.1 (Farey adjacency) 4

3 Farey Sequences **5**

4 Sturmian Sequences **5**
Definition 4.1 (Sturmian word) 5

5 Christoffel Words **5**
Definition 5.1 (Christoffel words) 5
Theorem 5.2 (Christoffel concatenation (Berstel [5], Theorem 1.5.1)) 6

II The Algorithm **6**

6 Definitions and Notation **6**
Definition 6.2 (Ratio) 7
Definition 6.3 (Mediant) 7
Definition 6.4 (Good estimate) 7
Definition 6.5 (Bounded pair) 7
Definition 6.6 (λ and ε) 7
Definition 6.7 (Δ and K) 8

7 The Mediant Construction **8**
Theorem 7.1 (Mediant convergence) 8
Corollary 7.2 (Straddling inequality) 9

8 The Optimal Weight Formula **9**
Theorem 8.1 (Optimal weight) 10
Corollary 8.2 (Children at w^+ form a bounded pair) 10

9 Mediant Words and the Mediant Tree **12**
9.1 Leaf words 12
Definition 9.1 (Leaf mediant word) 12
9.2 Branch words and the weighted operators 12
Definition 9.2 (Branch mediant word) 12
9.3 Mediant words are Christoffel words 13
Lemma 9.3 (Every mediant word is a lower Christoffel word) 13
9.4 The mediant tree 14
Definition 9.5 (Mediant tree) 14
9.5 The mediant word encodes the fraction 14

10 Convergence Analysis **15**
Theorem 10.1 (Strict decrease of $|\varepsilon|$) 15
Corollary 10.2 (Convergence to θ) 15

11 Examples and Computational Results **16**

12 Relationship to Sturmian Sequences **16**
12.1 Structural properties of mediant words 16
Theorem 12.1 (Every mediant word ends in O) 16
Theorem 12.2 (Overestimate criterion) 16
Theorem 12.4 (Universal EE prefix) 17
Theorem 12.5 (Palindrome invariant) 17

DRAFT

Theorem 12.6 (Sturmian identity) 17
Corollary 12.7 (Mismatch characterisation) 18

13 Extension to $0 < \theta < 1$ **20**
13.1 Case $0 < \theta < \frac{1}{2}$ 20
13.2 Case $\frac{1}{2} < \theta < 1$ 21

14 Conclusion and Open Problems **21**

A Proof of Theorem 8.1 **23**

B Examples and Computational Results **24**
B.1 Worked example: mediant words for $\log_2 3$ 24
B.2 Approximating $\log_2 3$ 25
B.3 Approximating π 25

1 Introduction

The approximation of irrational numbers by rationals is a classical subject in number theory. The Stern–Brocot tree and Farey sequences provide a systematic account of best rational approximants, and the continued-fraction expansion of a real number yields its convergents — the fractions closest to the target with bounded denominator. Sturmian words, introduced by Morse and Hedlund [1], give a combinatorial counterpart: the infinite binary sequence $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ with slope $\alpha \in (0, 1)$ is the unique balanced aperiodic sequence with O-density α.

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The present paper makes four contributions to this classical picture.

1. **Optimal weight formula.** We introduce the *bounded-pair* framework, in which two Farey-adjacent rational estimates bracket θ from above and below. Each weighted mediant step is a run of w steps in the Stern–Brocot tree in the same direction — the classical construction (see Graham, Knuth and Patashnik [2]). What is new is Theorem 8.1: a closed-form expression for the unique *optimal* integer weight w^+ — the one whose mediants straddle θ — derived from a sign-change bracket on the linear error function, enabling efficient iteration with no redundant steps.

2. **Mediant word representation.** The mediant tree can be encoded as a string over $\{\mathsf{O}, \mathsf{E}\}$, the *mediant word* (Section 9). This representation connects rational approximation to combinatorics on words.

3. **Sturmian identity.** The mediant word produced by the mediant tree is identical to the Sturmian word with slope $1/(1 + \theta)$ (Theorem 12.6), agreeing at every position except possibly the last symbol of each finite prefix. The identification of Christoffel words with Sturmian prefixes is classical (Lothaire [3]); the novelty is the direct proof via the mediant construction and the precise last-letter criterion.

4. **Compressed transducer.** Each iterate produced by the algorithm is at least a Stern–Brocot semi-convergent — the best bracketing fraction achievable in one step from the current bounded pair. The mediant tree is therefore a *compressed transducer* for the Sturmian word: the approximation path (empirically $O(\log N)$ integer-arithmetic steps; a formal bound remains open) encodes the entire construction history, after which the N symbols are enumerated by a purely integer recursive traversal with no further real arithmetic.

The algorithm requires an oracle for θ — a means of evaluating $\lceil o\theta \rceil$ to sufficient precision at each step — but is otherwise entirely general. Given such an oracle, the mediant tree can represent *any* real number as a mediant word over $\{\mathsf{O}, \mathsf{E}\}$, making it a universal encoder from reals to binary words.

Organisation. Part I (Sections 2–5) collects background on the Stern–Brocot tree, Farey sequences, Sturmian words, and Christoffel words. Part II develops the algorithm and its theory: Section 6 introduces the core definitions; Section 7 develops the mediant construction and straddling inequality; Section 8 derives the optimal weight formula; Section 9 introduces mediant words; Section 9.4 describes the mediant tree; Section 10 analyses the rate of convergence; Section 11 presents examples and computational results; Section 12 establishes the connection to Sturmian words; Section 13 extends the construction to all real $\theta \notin \{0, 1\}$; Section 14 concludes with open questions. Appendix A contains the proof of Theorem 8.1.

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Part I
Background

2 The Stern–Brocot Tree and Mediants

The *Stern–Brocot tree* is a complete binary search tree containing every positive rational exactly once in lowest terms (see, e.g., Graham, Knuth, and Patashnik [2]). It is constructed by repeated *mediant* insertion: starting from the pair $(0/1, 1/0)$, the mediant of two adjacent fractions a/b and c/d is $(a+c)/(b+d)$, and inserting it between them yields two new adjacent pairs. The Stern–Brocot tree encodes continued-fraction expansions as paths from the root: a partial quotient of k corresponds to k steps in the same direction before the first reversal.

Definition 2.1 (Farey adjacency)**.** Two fractions a/b and c/d (in lowest terms, $a/b < c/d$) are *Farey adjacent* if

$$bc - ad = 1.$$

Equivalently, they are adjacent in some Farey sequence F_n, or they lie on adjacent branches of the Stern–Brocot tree. Farey adjacency is equivalent to the condition that no rational with denominator at most $\max(b, d)$ lies strictly between a/b and c/d.

The mediant $(a + c)/(b + d)$ of two Farey-adjacent fractions is itself Farey adjacent to each of them. This adjacency-preservation is the key structural property exploited by our algorithm.

3 Farey Sequences

The *Farey sequence* F_n is the increasing sequence of fractions in $[0,1]$ with denominator at most n, arranged in order. Its fundamental property is that consecutive terms $a/b < c/d$ in F_n satisfy $bc - ad = 1$ (i.e. they are Farey adjacent), and their mediant $(a+c)/(b+d)$ is the fraction with smallest denominator strictly between them.

For our purposes the key fact is the following: if a/b and c/d are Farey adjacent and $\theta \in (a/b, c/d)$, then every rational approximant to θ with denominator at most $b+d$ is one of a/b, c/d, or their mediant $(a+c)/(b+d)$. This is why maintaining Farey adjacency throughout the algorithm ensures that no improvement step is redundant.

4 Sturmian Sequences

Sturmian words were introduced by Morse and Hedlund [1] as the simplest aperiodic symbolic sequences. Let $\alpha \in (0,1)$ be irrational. The *lower Sturmian word* with slope α is defined by

$$s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor, \qquad n \geq 0.$$

Encoding $s_n = 1$ as O and $s_n = 0$ as E, this is a binary sequence with O-density exactly α.

Definition 4.1 (Sturmian word)**.** The *Sturmian word* $\mathbf{s}(\alpha)$ with slope $\alpha \in (0,1)$ is the infinite word over $\{\mathsf{O}, \mathsf{E}\}$ defined by $s_n = \mathsf{O}$ if $\lfloor (n+1)\alpha \rfloor > \lfloor n\alpha \rfloor$, and $s_n = \mathsf{E}$ otherwise.

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Sturmian words are *balanced*: no two factors of the same length differ in their O-count by more than one. They are the unique balanced aperiodic sequences [1].

The two-word structure is fundamental: a Sturmian word with slope $\alpha \in (0, 1/2)$ consists of exactly two distinct *primitive words* (unbreakable building blocks), of lengths $\lfloor 1/\alpha \rfloor$ and $\lceil 1/\alpha \rceil$, concatenated in the Sturmian pattern. Concretely, with $\alpha = 1/(1+\theta)$ for $\theta > 1$, the two primitive words are $\mathsf{E}^{\lfloor \theta \rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil \theta \rceil}\mathsf{O}$ — exactly the leaf words of the mediant tree (Section 9.4).

Since $\alpha < 1/2$ for $\theta > 1$, the sequence avoids OO: two consecutive O's would require $\lfloor (n+2)\alpha \rfloor - \lfloor n\alpha \rfloor = 2$, but $2\alpha < 1$ makes this impossible. Conversely, when $\alpha > 1/2$ (i.e. $\theta < 1$), consecutive O's are unavoidable.

Sturmian morphisms. The two *Sturmian morphisms* $\sigma_E \colon \mathsf{E} \mapsto \mathsf{E},\ \mathsf{O} \mapsto \mathsf{EO}$ and $\sigma_O \colon \mathsf{E} \mapsto \mathsf{OE},\ \mathsf{O} \mapsto \mathsf{O}$ map Sturmian words to Sturmian words (of a different slope in general).

5 Christoffel Words

Definition 5.1 (Christoffel words)**.** For coprime positive integers e and o, the *lower Christoffel word* $\mathrm{Ch}^-(e,o)$ is the unique primitive word of length $L = e + o$ over $\{\mathsf{E}, \mathsf{O}\}$ containing exactly e copies of E and o copies of O that ends in O, with O-positions given by

$$\mathrm{Ch}^-(e,o)[i] = \mathsf{O} \iff \lfloor (i+1)o/L \rfloor > \lfloor i \cdot o/L \rfloor.$$

The *upper Christoffel word* $\mathrm{Ch}^+(e,o)$ is the reverse of $\mathrm{Ch}^-(e,o)$: it ends in E instead of O. Both words share the same palindromic interior U: $\mathrm{Ch}^-(e,o) = \mathsf{E}\cdot U\cdot\mathsf{O}$ and $\mathrm{Ch}^+(e,o) = \mathsf{O}\cdot U\cdot\mathsf{E}$ (Berstel [5], Chapter 1).

$\mathrm{Ch}^-(e,o)$ is the lower Sturmian word of slope o/L truncated to length L and normalised to end in O; $\mathrm{Ch}^+(e,o)$ is the upper Sturmian word of the same slope, ending in E (Berstel [5], Chapter 1).

Each Christoffel word is characterised purely by its fraction o/L: given $\gcd(e,o)=1$, the pair (e,o) is uniquely determined by $o/(e+o)$.

Theorem 5.2 (Christoffel concatenation (Berstel [5], Theorem 1.5.1))**.** *Let* $\mathrm{Ch}^-(e_0,o_0)$ *and* $\mathrm{Ch}^-(e_1,o_1)$ *be two lower Christoffel words with* $o_0/L_0 < o_1/L_1$ *(where* $L_k = e_k+o_k$*) satisfying the Farey adjacency condition* $o_1L_0 - o_0L_1 = 1$*. Then for any integer* $w \geq 1$*:*

$$\begin{aligned}\mathrm{Ch}^-(e_0,o_0)^w\cdot\mathrm{Ch}^-(e_1,o_1) &= \mathrm{Ch}^-(w\,e_0+e_1,\ w\,o_0+o_1),\\ \mathrm{Ch}^-(e_0,o_0)\cdot\mathrm{Ch}^-(e_1,o_1)^w &= \mathrm{Ch}^-(e_0+w\,e_1,\ o_0+w\,o_1).\end{aligned}$$

In the first identity the smaller-slope word is repeated; in the second the larger-slope word is repeated.

Part II
The Algorithm

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6 Definitions and Notation

Throughout Sections 6–12, let $\theta > 1$ be a fixed real number. Section 13 handles all other real numbers by reduction to this case. We write $\mathbb{N}$ for the positive integers and $\lfloor\cdot\rfloor$, $\lceil\cdot\rceil$ for floor and ceiling.

Remark 6.1 (Rational vs. irrational θ). Most of the combinatorial claims in these sections — the Christoffel word structure (Lemma 9.3), the palindrome invariant (Theorem 12.5), and the EE-prefix theorem (Theorem 12.4) — hold for all real $\theta > 1$, rational or not. The Sturmian identity (Theorem 12.6) and the last-letter theorem (Theorem 12.2) require θ to be irrational, since the Sturmian word is only well-defined for irrational slope $\alpha = 1/(1+\theta)$. The uniqueness of w^+ (Theorem 8.1) also requires irrationality. The formula $w^+ = \lfloor -A/B\rfloor + 1$ rests on $-A/B$ being non-integer: the strict bracketing condition $(A+(w-1)B)(A+wB) < 0$ holds for exactly one integer w iff $-A/B$ is not an integer. For rational θ, $-A/B$ can be an integer w^-, in which case no integer satisfies the strict condition: weights w^- and $w^+ = w^-+1$ give product zero, and all others give a positive product. The mediant at weight w^- equals θ exactly. Step 4 of the algorithm then computes $w^+ = w^- + 1$, forming a pair whose first member is θ exactly; $K = 0$ is detected at the next pass through step 3, at which point the algorithm terminates. For rational θ the algorithm terminates in finitely many steps (at the step where $K = 0$, i.e. where r_{2d+1} is an exact representation of θ); all claims except Theorems 12.2, 12.6 and Corollary 12.7 hold for every iterate up to termination.

Definition 6.2 (Ratio)**.** A *ratio* is a 2-tuple $r = (e, o) \in \mathbb{N}^2$ with $\gcd(e, o) = 1$, representing the rational number e/o. A ratio $r = (e, o)$ is a *rational estimate* of θ when used in that context.

Definition 6.3 (Mediant)**.** Given two ratios $r = (e, o)$ and $r' = (e', o')$ and a non-negative integer $w \geq 0$, the *mediant* of r and r' with weight w is the 3-tuple

$$\text{Mediant}(r,\, r',\, w) = (r,\, r',\, w),$$

treated as a ratio by the composition rule

$$\text{Mediant}(r,\, r',\, w) = (e + w\, e',\ o + w\, o').$$

The degenerate case $w = 0$ gives $\text{Mediant}(r, r', 0) = r$. The Farey determinant of the pair (r, r') is

$$\Delta(r, r') = e'o - eo'.$$

Definition 6.4 (Good estimate)**.** An estimate e/o is a *good estimate* of θ if

$$e \in \{\lfloor o \cdot \theta \rfloor,\, \lceil o \cdot \theta \rceil\},$$

i.e. e/o is one of the two nearest rationals to θ with denominator o. A good estimate with $e = \lceil o\theta \rceil$ satisfies $e/o \geq \theta$ and is an *upper bound*; one with $e = \lfloor o\theta \rfloor$ satisfies $e/o \leq \theta$ and is a *lower bound*.

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Definition 6.5 (Bounded pair)**.** Two good estimates e_l/o_l and e_u/o_u form a *bounded pair* if e_l/o_l is a lower bound, e_u/o_u is an upper bound,

$$\frac{e_l}{o_l} < \theta \leq \frac{e_u}{o_u},$$

and they satisfy the *Farey adjacency condition*

$$e_u \cdot o_l - e_l \cdot o_u = 1.$$

The Farey adjacency condition ensures that the pair lies on adjacent branches of the Stern–Brocot tree (Section 2) and that no rational with smaller denominator lies strictly between e_l/o_l and e_u/o_u.

Definition 6.6 (λ and ε)**.** For an estimate e/o of θ, define the *relative ratio*

$$\lambda = \frac{e/o}{\theta}$$

and the *relative error*

$$\varepsilon = \lambda - 1 = \frac{e/o}{\theta} - 1.$$

An upper-bound good estimate satisfies $\lambda \geq 1$ (so $\varepsilon \geq 0$); a lower-bound good estimate satisfies $\lambda \leq 1$ (so $\varepsilon \leq 0$). In both cases $|\varepsilon|$ is small when o is large.

Estimates are indexed so that r_{2d} and r_{2d+1} form the bounded pair at depth d, with r_{2d+1} always the better estimate (smaller $|\varepsilon|$). Their children are r_{2d+2} and r_{2d+3}, the bounded pair at depth $d+1$. We write λ_{2d}, ε_{2d}, etc. for the corresponding values. In a bounded pair we always orient so that $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$, i.e. r_{2d+1} is the better estimate of the two.

Definition 6.7 (Δ and K)**.** Given a bounded pair $(r_{2d}, r_{2d+1}) = (e_{2d}/o_{2d},\, e_{2d+1}/o_{2d+1})$ with $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$, define

$$\Delta = e_{2d+1}o_{2d} - e_{2d}o_{2d+1}$$

and

$$K = o_{2d+1}(e_{2d+1} - \theta \cdot o_{2d+1}).$$

Geometrically, $K = \theta\, o_{2d+1}^2 \varepsilon_{2d+1}$ is the signed absolute error of the better estimate r_{2d+1} scaled by its denominator squared: it measures how far r_{2d+1} sits from the target line $e = \theta o$ in units normalised to the lattice cell size.

The Farey adjacency condition in Definition 6.5 requires $e_u \cdot o_l - e_l \cdot o_u = 1$, which is precisely $|\Delta| = 1$ (the sign depends on which of r_{2d}, r_{2d+1} is the upper bound). One checks that $|K|/o_{2d+1} = |e_{2d+1} - \theta o_{2d+1}| < 1$ (the good-estimate condition), and that $\operatorname{sgn}(K) = \operatorname{sgn}(\Delta)$ throughout the iteration.

Domain restriction. The definitions in this section assume $\theta > 1$. Section 13 shows how to reduce all other real numbers to this case.

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Bounded pairs and the mediant tree. A bounded pair is a purely analytic object: any two good estimates that bracket θ and satisfy the Farey adjacency condition qualify, regardless of how they were found. The mediant construction (Section 7) provides a canonical way to *generate* bounded pairs, but the concept is wider. The invariants Δ and K belong to the pair itself; the Christoffel word structure (Section 5) is a property of how the pair was reached via the mediant tree. In particular, $\Delta = \pm 1$ and $\operatorname{sgn}(K) = \operatorname{sgn}(\Delta)$ hold for any bounded pair, while the palindromic interior of $\pi(r)$ is a certificate that the estimate r was produced by the mediant algorithm.

7 The Mediant Construction

The classical *mediant* of two fractions e_{2d}/o_{2d} and e_{2d+1}/o_{2d+1} is $(e_{2d} + e_{2d+1})/(o_{2d} + o_{2d+1})$. We generalise this to a weighted family parametrised by a positive integer w.

Theorem 7.1 (Mediant convergence)**.** *Let (r_{2d}, r_{2d+1}) be a bounded pair for θ with $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$, and let $w \geq 1$ be a positive integer such that* $(\text{Mediant}(r_{2d}, r_{2d+1}, w{-}1),\ \text{Mediant}(r_{2d}, r_{2d+1}, w))$ *brackets θ. Write $r_- = \text{Mediant}(r_{2d}, r_{2d+1}, w-1)$ and $r_+ = \text{Mediant}(r_{2d}, r_{2d+1}, w)$ for these two candidates. Then*

$$|\varepsilon_-| \leq |\varepsilon_{2d}| \qquad \textit{and} \qquad |\varepsilon_+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|.$$

(When $w = 1$, $r_- = r_{2d}$ exactly so $|\varepsilon_-| = |\varepsilon_{2d}|$; otherwise $|\varepsilon_-| < |\varepsilon_{2d}|$.) Moreover $|e_+\, o_- - e_-\, o_+| = 1$, so (r_-, r_+) is Farey-adjacent. The two mediants are assigned as r_{2d+2} and

r_{2d+3}, where r_{2d+3} is whichever has the smaller $|\varepsilon|$ (the better estimate), and r_{2d+2} takes the other. Since r_+ always satisfies $|\varepsilon_+| < |\varepsilon_{2d+1}|$, we have $|\varepsilon_{2d+3}| < |\varepsilon_{2d+1}|$ regardless of which candidate is assigned to r_{2d+3}.

Proof. The denominators satisfy $o_- = o_{2d} + (w-1)o_{2d+1} < o_{2d} + w\, o_{2d+1} = o_+$, so r_- and r_+ are distinct. The ratio λ_- is a weighted combination of λ_{2d} and λ_{2d+1}:

$$\lambda_- = \frac{o_{2d}\,\lambda_{2d} + (w-1)o_{2d+1}\,\lambda_{2d+1}}{o_{2d} + (w-1)o_{2d+1}}.$$

When $w = 1$ this gives $\lambda_- = \lambda_{2d}$ exactly, so $|\varepsilon_-| = |\varepsilon_{2d}|$. When $w \geq 2$, r_- lies strictly between r_{2d} and r_{2d+1}. Write $f(t) = A + tB$ where $A = e_{2d} - \theta o_{2d}$ and $B = e_{2d+1} - \theta o_{2d+1}$ are the scaled errors of r_{2d} and r_{2d+1}; then $f(w) = e_+ - \theta\, o_+$. From the bracketing hypothesis $f(w-1)f(w) < 0$ it follows that $B \neq 0$ (otherwise f would be constant and could not change sign), and that the zero $c = -A/B$ of f lies in $(w-1, w)$. Thus $f(w-1)$ has the sign of A (the side of r_{2d}) and $f(w)$ has the sign of B (the side of r_{2d+1}). Since the bracketing condition places r_- on the same side of θ as r_{2d}, the ratio r_-/θ lies between r_{2d}/θ and 1, so $|\lambda_- - 1| < |\lambda_{2d} - 1|$, i.e. $|\varepsilon_-| < |\varepsilon_{2d}|$. For r_+ (weight $w \geq 1$), the bracketing condition places r_+ on the same side of θ as r_{2d+1}, so r_+/θ lies between r_{2d+1}/θ and 1, giving $|\lambda_+ - 1| < |\lambda_{2d+1} - 1|$, i.e. $|\varepsilon_+| < |\varepsilon_{2d+1}| < |\varepsilon_{2d}|$. The Farey adjacency condition for (r_-, r_+) follows by direct expansion:

$$\begin{aligned} e_+\, o_- - e_-\, o_+ &= (e_{2d} + w\, e_{2d+1})(o_{2d} + (w-1)o_{2d+1}) - (e_{2d} + (w-1)e_{2d+1})(o_{2d} + w\, o_{2d+1}) \\ &= \big[(w-1) - w\big] e_{2d}\, o_{2d+1} + \big[w - (w-1)\big] e_{2d+1}\, o_{2d} \\ &= -e_{2d}\, o_{2d+1} + e_{2d+1}\, o_{2d} = \Delta, \end{aligned}$$

so $|e_+\, o_- - e_-\, o_+| = |\Delta| = 1$, and the Farey adjacency of (r_-, r_+) is inherited directly from that of (r_{2d}, r_{2d+1}). □

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Corollary 7.2 (Straddling inequality)**.** *Under the conditions of Theorem 7.1,*

$$\varepsilon_-\,\varepsilon_+ < 0 \qquad \textit{and} \qquad |\varepsilon_-||\varepsilon_+| < |\varepsilon_{2d}|\,|\varepsilon_{2d+1}|.$$

Proof. The first inequality holds because r_- and r_+ straddle θ, so their errors have opposite signs. For the second: $|\varepsilon_-| \leq |\varepsilon_{2d}|$ and $|\varepsilon_+| < |\varepsilon_{2d+1}|$ (Theorem 7.1), so $|\varepsilon_-||\varepsilon_+| \leq |\varepsilon_{2d}||\varepsilon_+| < |\varepsilon_{2d}|\,|\varepsilon_{2d+1}|$. □

8 The Optimal Weight Formula

A key question is: for which integer $w \geq 1$ do the two consecutive weighted mediants

$$r_- = \text{Mediant}(r_{2d}, r_{2d+1}, w-1), \qquad r_+ = \text{Mediant}(r_{2d}, r_{2d+1}, w)$$

straddle θ, i.e. satisfy the bracketing condition $(\lambda_- - 1)(\lambda_+ - 1) < 0$?

Writing $A = e_{2d} - \theta\, o_{2d}$ and $B = e_{2d+1} - \theta\, o_{2d+1}$ for the scaled errors, the bracketing condition becomes $(A + (w-1)B)(A + wB) < 0$. Observe that $A + (w-1)B$ and $A + wB$

are consecutive evaluations of the linear function $f(w) = A + wB$ at $w-1$ and w: their product is negative iff f changes sign between $w-1$ and w, i.e. iff the zero $-A/B$ of f lies in the interval $(w-1, w)$, i.e. iff $w = \lfloor -A/B \rfloor + 1$. Since θ is irrational, $-A/B$ is never an integer, so exactly one such w exists. The unique integer satisfying this bracketing condition is given by the following closed-form expression.

Theorem 8.1 (Optimal weight)**.** *Let (r_{2d}, r_{2d+1}) be a bounded pair bracketing θ, with*

$$A = e_{2d} - \theta\, o_{2d}, \qquad B = e_{2d+1} - \theta\, o_{2d+1},$$

and set $\Delta = e_{2d+1}o_{2d} - e_{2d}o_{2d+1}$, $K = o_{2d+1}B$. Then the unique integer $w \geq 1$ for which $r_- = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w-1)$ and $r_+ = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w)$ satisfy the bracketing condition $(\lambda_- - 1)(\lambda_+ - 1) < 0$ is

$$w^+ = \left\lfloor \frac{-A}{B} \right\rfloor + 1 = \left\lfloor \frac{\Delta}{K} - \frac{o_{2d}}{o_{2d+1}} \right\rfloor + 1.$$

A derivation is given in Appendix A.

When $K = 0$, r_{2d+1} is an exact rational representation of θ and no further improvement is possible; the algorithm terminates.

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Corollary 8.2 (Children at w^+ form a bounded pair)**.** *Under the conditions of Theorem 8.1, the two children $r_- = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^-)$ and $r_+ = \mathrm{Mediant}(r_{2d}, r_{2d+1}, w^+)$ form a bounded pair.*

Proof. Farey adjacency $|e_+ o_- - e_- o_+| = 1$ follows from Theorem 7.1. It remains to show both children are good estimates. Write $f(w) = A + wB$ so that $f(w) = e_w - \theta o_w$ (the scaled error of r_w). The bracketing condition $f(w^-)\, f(w^+) < 0$ means the two values have opposite signs, so

$$|f(w^-)|\, o_+ + |f(w^+)|\, o_- = |f(w^-)\, o_+ - f(w^+)\, o_-| = |e_-\, o_+ - e_+\, o_-| = |\Delta| = 1.$$

Both terms on the left are in fact positive: since θ is irrational (Remark 6.1), $f(w) \neq 0$ for all integers w, so $|f(w^-)|$ and $|f(w^+)|$ are both nonzero; and $o_-, o_+ \geq 1$. Since two positive terms sum to 1, each is strictly less than 1, i.e. $|e_- - \theta o_-| < 1$ and $|e_+ - \theta o_+| < 1$, confirming both children are good estimates. Moreover, $|\Delta| = 1$ directly gives $\gcd(e_+, o_+) = \gcd(e_-, o_-) = 1$, so the children are ratios in lowest terms. □

The continuous optimum and the discrete formula coincide

To see why w^+ is the natural choice, consider treating w as a real variable and finding the stationary point of the error product $f(w-1)\, f(w)$, where $f(w) = A + wB$. Differentiating with respect to w and setting the derivative to zero:

$$\frac{d}{dw}\big[f(w-1)\, f(w)\big] = B\big[(A + wB) + (A + (w-1)B)\big] = 0.$$

Solving gives the unique real critical point

$$w^* = \tfrac{1}{2} - \frac{A}{B} = c + \tfrac{1}{2},$$

where $c = -A/B > 0$. The product $f(w-1)\,f(w)$ has roots at $w = c$ (where $f(w) = 0$) and $w = c+1$ (where $f(w-1) = 0$), and $w^* = c + \frac{1}{2}$ is their midpoint — the unique real maximum of $|f(w-1)\,f(w)|$ between the roots. The unique integer w for which $f(w-1)$ and $f(w)$ have opposite signs satisfies $c < w < c+1$, i.e. $w = \lfloor c \rfloor + 1 = w^+$. Since $w^* = c + \frac{1}{2}$ also lies in $(c,\, c+1)$, and w^+ is the unique integer in that interval, w^+ is the nearest integer to w^*: both the continuous stationary point and the sign-change bracket single out the same w^+.

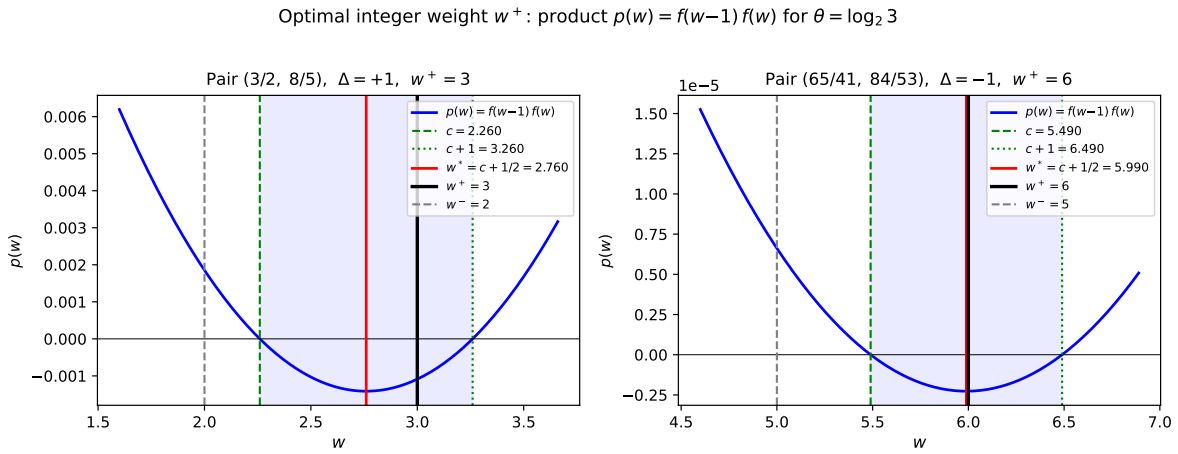


Figure 1: The product $p(w) = f(w-1)\,f(w)$ for two pairs from the mediant tree for $\theta = \log_2 3$, illustrating the optimal integer weight w^+. *Left:* pair (3/2, 8/5) with $\Delta = +1$, $c \approx 2.260$, and $w^+ = 3$. *Right:* pair (65/41, 84/53) with $\Delta = -1$, $c \approx 5.490$, and $w^+ = 6$. In each case the roots of p are at c (where $f(w) = 0$, green dashed) and $c+1$ (where $f(w-1) = 0$, green dotted); the continuous stationary point $w^* = c + \frac{1}{2}$ (red) is their midpoint; the unique integer in the shaded interval $(c,\, c+1)$ is w^+ (black); and its companion $w^- = w^+ - 1$ (grey dashed) lies just outside the interval on the left. Together w^- and w^+ are the two weights whose mediants bracket θ. The two panels illustrate both signs of Δ.

Algorithm. The iteration proceeds as follows.

1. Start with an initial bounded pair (r_0, r_1) for θ.
2. Compute Δ and K.
3. If $K = 0$, terminate.
4. Compute $w^+ = \lfloor \Delta/K - o_{2d}/o_{2d+1} \rfloor + 1$ and set $w^- = w^+ - 1$.
5. Form the new bounded pair $(\text{Mediant}(r_{2d}, r_{2d+1}, w^-),\ \text{Mediant}(r_{2d}, r_{2d+1}, w^+))$ and repeat from step 2.

Because o_{2d+1} grows monotonically along the iteration, $|\varepsilon|$ shrinks and the iteration converges to θ (Corollary 10.2).

9 Mediant Words and the Mediant Tree

The mediant construction admits a compact combinatorial encoding. Every estimate produced by the algorithm is either a *leaf* or a *branch*, and the recursive structure of all estimates defines a binary tree whose nodes carry both rational values and combinatorial words over $\{\mathsf{O}, \mathsf{E}\}$.

9.1 Leaf words

Definition 9.1 (Leaf mediant word)**.** Let $\{\mathsf{O}, \mathsf{E}\}^*$ denote the set of finite strings over the two-letter alphabet $\{\mathsf{O}, \mathsf{E}\}$. A *leaf* estimate has denominator $o = 1$, i.e. it is a base estimate $e/1$ for some positive integer e. Its *mediant word* is

$$\pi(e/1) = \underbrace{\mathsf{E} \cdots \mathsf{E}}_{e} \mathsf{O},$$

a string of e copies of E followed by one O.

The two leaf estimates that initialise the algorithm are $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$, ordered as (r_0, r_1) so that r_1 is the better estimate (i.e. $|\varepsilon_{r_1}| < |\varepsilon_{r_0}|$). Their mediant words are $\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil\theta\rceil}\mathsf{O}$ respectively. Since θ is irrational, $\lceil\theta\rceil = \lfloor\theta\rfloor + 1$, so $\pi(r_0)$ and $\pi(r_1)$ differ by exactly one E. Each word $\mathsf{E}^k\mathsf{O}$ contains exactly one O, so it is not a power of a shorter word; both $\pi(r_0)$ and $\pi(r_1)$ are therefore primitive. Moreover, $\mathsf{E}^k\mathsf{O}$ is the lower Christoffel word $\mathrm{Ch}^-(k, 1)$ (Definition 5.1): its O appears at position k, matching the Christoffel formula. Thus both leaf words are lower Christoffel words of their respective fractions.

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9.2 Branch words and the weighted operators

Each branch step produces two new mediant words from the current pair $(u, v) = (\pi(r_{2d}), \pi(r_{2d+1}))$ by concatenation:

Definition 9.2 (Branch mediant word)**.** A *branch* estimate r_{2d+2} or r_{2d+3} has two *ordered parents* (r_{2d}, r_{2d+1}) and a positive integer *weight* w. Estimates are indexed so that r_{2d} and r_{2d+1} form the bounded pair at depth d, with r_{2d+1} always the better estimate (smaller $|\varepsilon|$). The optimal weight w^+ computed from (r_{2d}, r_{2d+1}) equals w_{2d+1} in this indexing; the two child estimates r_{2d+2} and r_{2d+3} carry weights w^- and w^+ (i.e. w_{2d+2} and w_{2d+3} are $w^+ - 1$ and w^+ in some order). Write $u = \pi(r_{2d})$ and $v = \pi(r_{2d+1})$ for the parent mediant words.

The two branch words at each depth share the same parents and differ only in the multiplicity of v: one estimate uses weight w^+ and the other uses weight w^-. Which is the better estimate depends on the errors: writing ε_- and ε_+ for the errors of the two candidates (Definition 6.6),

$$w_{2d+3} - w_{2d+2} = \begin{cases} +1 & \text{if } |\varepsilon_+| < |\varepsilon_-|, \\ -1 & \text{otherwise.} \end{cases}$$

In either case $|\pi(r_{2d+3})| - |\pi(r_{2d+2})| = \pm|v|$.

The branch mediant words are:

- $\Delta > 0$ (r_{2d+1} is the upper bound, v plays the role of E):

$$\pi(r_{2d+2}) = v^{w-1} \cdot u,$$
$$\pi(r_{2d+3}) = v^{w} \cdot u.$$

- $\Delta < 0$ (r_{2d+1} is the lower bound, v plays the role of O):

$$\pi(r_{2d+2}) = u \cdot v^{w-1},$$
$$\pi(r_{2d+3}) = u \cdot v^{w}.$$

Which of the two words is the better estimate r_{2d+3} is determined by comparing $|\varepsilon_-|$ and $|\varepsilon_+|$; the assignment is not fixed in advance.

Concrete example ($w = 2$, depth-0 leaves $\pi(r_0) = \mathtt{EO}$, $\pi(r_1) = \mathtt{EEO}$, $\Delta > 0$). Write $\Phi^w(v, u) = (v^w \cdot u,\ v^{w-1} \cdot u)$ for the pair of branch words at weight w (the $\Delta > 0$ case):

$$\Phi^2\big(\mathtt{EEO},\, \mathtt{EO}\big) = \big(\mathtt{EEO}^2\,\mathtt{EO},\ \mathtt{EEO}\,\mathtt{EO}\big) = \big(\mathtt{EEOEEOEO},\ \mathtt{EEOEO}\big),$$

so $\pi(r_3) = \mathtt{EEOEEOEO}$ and $\pi(r_2) = \mathtt{EEOEO}$ (the second component of $\Phi^1(\mathtt{EEO}, \mathtt{EO})$, the $w{-}1$ companion pair).

DRAFT

9.3 Mediant words are Christoffel words

A word is *primitive* if it is not a power x^k of a strictly shorter word x (with $k \geq 2$).

Lemma 9.3 (Every mediant word is a lower Christoffel word)**.** *For every estimate $r = (e, o)$ produced by the mediant tree, $\pi(r) = \mathrm{Ch}^-(e, o)$, the lower Christoffel word at slope $o/(e+o)$. In particular, $\pi(r)$ is never an upper Christoffel word, since by Theorem 12.1 every mediant word ends in O, whereas upper Christoffel words end in E.*

Proof. By structural induction on the mediant tree.

Base case. By §9.1, $\pi(e/1) = \mathsf{E}^e\mathsf{O} = \mathrm{Ch}^-(e, 1)$.

Inductive step. Let (r_{2d}, r_{2d+1}) be a bounded pair with $\Delta = e_{2d+1}o_{2d} - e_{2d}o_{2d+1}$. By induction $\pi(r_{2d}) = \mathrm{Ch}^-(e_{2d}, o_{2d})$ and $\pi(r_{2d+1}) = \mathrm{Ch}^-(e_{2d+1}, o_{2d+1})$, both lower Christoffel words (ending in O), with slopes o_{2d}/L_{2d} and o_{2d+1}/L_{2d+1} (where $L_k = e_k + o_k$) on opposite sides of $\alpha = 1/(1+\theta)$.

When $\Delta > 0$: r_{2d+1} is the upper estimate so $e_{2d+1}/o_{2d+1} > \theta$, giving $o_{2d+1}/L_{2d+1} < \alpha < o_{2d}/L_{2d}$. The new word is $v^w u = \pi(r_{2d+1})^w \cdot \pi(r_{2d})$: the smaller-slope word $\pi(r_{2d+1})$ repeated w times, followed by $\pi(r_{2d})$ once. By Theorem 5.2 (with $e_0 = e_{2d+1}$, $e_1 = e_{2d}$) this equals $\mathrm{Ch}^-(w\,e_{2d+1} + e_{2d},\ w\,o_{2d+1} + o_{2d})$.

When $\Delta < 0$: r_{2d+1} is the lower estimate so $e_{2d+1}/o_{2d+1} < \theta$, giving $o_{2d}/L_{2d} < \alpha < o_{2d+1}/L_{2d+1}$. Thus $\pi(r_{2d})$ has the smaller slope and $\pi(r_{2d+1})$ has the larger slope. The new word is $uv^w = \pi(r_{2d}) \cdot \pi(r_{2d+1})^w$: the single smaller-slope word followed by the larger-slope word repeated w times. By Theorem 5.2 (with $e_0 = e_{2d}$, $e_1 = e_{2d+1}$, $o_0/L_0 < o_1/L_1$, Berstel [5] Theorem 1.5.1) this equals $\mathrm{Ch}^-(e_{2d} + w\,e_{2d+1},\ o_{2d} + w\,o_{2d+1})$.

In both cases $\gcd(e_{2d} + we_{2d+1},\ o_{2d} + wo_{2d+1}) = 1$ by the Farey adjacency condition $|e_{2d+1}o_{2d} - e_{2d}o_{2d+1}| = 1$, confirming the result is a valid lower Christoffel word. □

The Christoffel identification is the key fact connecting the mediant algorithm to the combinatorics of Sturmian words. The recurrence $\pi(r_{2d+3}) = v^w \cdot u$ (or $u \cdot v^w$) matches the standard sequence generation rule for Sturmian words (Widmer [4], Proposition 2.4.6), with weight w encoding the partial quotient at that depth.

Remark 9.4. The Christoffel tree — the binary tree whose nodes are lower Christoffel words under the same concatenation rules — is a classical object (Berstel et al. [5], Chapter 2). Lemma 9.3 says that the mediant algorithm traverses exactly this tree: every estimate produced is a node of the Christoffel tree, and the branch rule is the Christoffel concatenation. The novelty is not the tree itself but the *driving criterion*: the optimal weight w^+ (Theorem 8.1) selects the unique child that straddles θ, making the traversal goal-directed rather than exhaustive.

9.4 The mediant tree

The mediant tree is the practical device for generating mediant words of arbitrary length and precision. Given a target θ and a desired precision, one runs the algorithm until the bounded pair (r_{2d}, r_{2d+1}) achieves the required length $e_{2d+1}+o_{2d+1}$. The mediant word $\pi(r_{2d+1})$ is then read off the tree by the recursive rule: an interior node $\text{Mediant}(r_0, r_1, w)$ with $\Delta > 0$ yields $\pi(r_1)^w \cdot \pi(r_0)$, and with $\Delta < 0$ yields $\pi(r_0) \cdot \pi(r_1)^w$. Deeper trees produce longer prefixes of the Sturmian word with slope $1/(\theta + 1)$ (Section 12).

DRAFT

Definition 9.5 (Mediant tree)**.** The *mediant tree* of an estimate r is defined recursively:

- If $r \in \{\lfloor\theta\rfloor, \lceil\theta\rceil\}$, then r is a *leaf node*.
- Otherwise r is an *interior node* $\text{Mediant}(r_0, r_1, w)$ with left subtree r_0, right subtree r_1, and integer weight $w \geq 0$, representing the rational

$$r = \frac{e_0 + w\, e_1}{o_0 + w\, o_1}.$$

The two leaves are the unique consecutive integers bracketing θ, forming the initial bounded pair (r_0, r_1), ordered so that r_1 is the better estimate (i.e. $|\varepsilon_{r_1}| < |\varepsilon_{r_0}|$). Each bounded pair (r_{2d}, r_{2d+1}) in the algorithm's sequence consists of two mediant trees sharing the same leaves; as the iteration advances the trees grow deeper, recording the full construction history of each estimate.

A detailed worked example for $\theta = \log_2 3$, tracing the first four steps of the algorithm and the corresponding mediant word decompositions, is given in Appendix B.

9.5 The mediant word encodes the fraction

By Lemma 9.3, the mediant word $\pi(r)$ of an estimate $r = e/o$ is uniquely determined by its fraction: $\pi(r) = \text{Ch}^-(e, o)$. The fraction e/o records the full construction history of r via the Euclidean structure of the mediant algorithm. Conversely, the fraction e/o can be read off from $\pi(r)$ by counting E's and O's. As the iteration advances, the ratio of E's to O's in $\pi(r_{2d+1})$ converges to θ, with strict improvement guaranteed across iterations (Theorem 10.1) but not within a single layer.

10 Convergence Analysis

Theorem 10.1 (Strict decrease of $|\varepsilon|$)**.** *At each iteration of the algorithm, $|\varepsilon_{2d+1}|$ strictly decreases.*

Proof. By Theorem 7.1, $|\varepsilon_+| < |\varepsilon_{2d+1}|$. The next iterate assigns the better of r_- and r_+ as r_{2d+3}, so $|\varepsilon_{2d+3}| \le |\varepsilon_+| < |\varepsilon_{2d+1}|$. □

Corollary 10.2 (Convergence to θ)**.** *The sequence of better estimates produced by the algorithm satisfies $\varepsilon_{2d+1} \to 0$.*

Proof. From the proof of Corollary 8.2, writing $f(w) = A + wB$ for the linear error function and o_+, o_- for the two child denominators:

$$|f(w^-)|\, o_+ + |f(w^+)|\, o_- = |\Delta| = 1.$$

Since $f(w) = e_w - \theta\, o_w$ and $\varepsilon_w = f(w)/(\theta\, o_w)$, substituting $|f(w)| = |\varepsilon_w|\, \theta\, o_w$ gives

$$\big(|\varepsilon_-| + |\varepsilon_+|\big)\, \theta\, o_-\, o_+ = 1,$$

so in particular the better child satisfies

$$|\varepsilon_{\text{better}}| \le \frac{1}{2\,\theta\, o_-\, o_+}.$$

DRAFT

(When $w^- = w^+ - 1 = 0$, the companion r_- equals the parent r_{2d}, which already satisfies $|\varepsilon_{2d}| > |\varepsilon_{2d+1}|$; the better child is then r_+, and the bound $|\varepsilon_+| \le 1/(2\theta\, o_+)$ still holds.) Since $o_+ = o_{2d} + w^+\, o_{2d+1} \ge o_{2d} + o_{2d+1} > o_{2d+1}$, the denominator of the better child strictly exceeds o_{2d+1}; by induction the denominators diverge. Since $|\varepsilon_{\text{better}}| \le 1/(2\,\theta\, o_-\, o_+)$ and $o_+ \to \infty$, we have $\varepsilon_{2d+1} \to 0$. □

Connection to continued fractions

The connection between the mediant algorithm and continued fractions is a direct consequence of the Sturmian identity (Theorem 12.6). In combinatorics on words, the standard sequence for a Sturmian word with slope $\alpha = 1/(1+\theta)$ is generated by the recurrence $s_n = s_{n-1}^{d_n} \cdot s_{n-2}$, where the weights d_n are precisely the partial quotients of the continued-fraction expansion of α (Lothaire [3], Theorem 2.3.5; Widmer [4], Proposition 2.4.6). The mediant word sequence agrees with this standard sequence at all interior positions (Theorem 12.6). This does *not* imply that the algorithm's weights w^+ equal the partial quotients a_k: the Sturmian identity constrains only interior positions of the words, not which weights the algorithm selects. What does follow is that the two children at weights w^- and w^+ always have denominators $o_{2d} + w^- \cdot o_{2d+1}$ and $o_{2d} + w^+ \cdot o_{2d+1}$ (one even-indexed, one odd-indexed depending on which is the better estimate), mirroring the structure of the classical recurrence $q_{k+1} = a_k q_k + q_{k-1}$. Whether this implies superlinear denominator growth remains open.

The MT fractions are not in general the same as the CF convergents. The standard CF recurrence jumps by the full partial quotient a_k at each step, landing on a new convergent p_k/q_k. The MT algorithm picks the optimal bracketing weight w^+, which may be strictly less than a_k: in that case the new estimate is a *semi-convergent* — a Stern-Brocot fraction between two consecutive CF convergents — not a CF convergent itself. Conversely, when w^+ is large it can jump past several intermediate semi-convergents in one step, skipping CF convergents that lie between them. Neither sequence is a subsequence of the other: the MT visits some semi-convergents that CF skips, and CF visits some convergents that the MT skips. The two sequences are related via the shared Sturmian word structure, but they traverse different fractions.

11 Examples and Computational Results

Extended tables illustrating the algorithm on $\theta = \log_2 3$ and $\theta = \pi$ are given in Appendix B.

12 Relationship to Sturmian Sequences

A *Sturmian word* over $\{\mathsf{O}, \mathsf{E}\}$ with slope $\alpha \in (0, 1)$ is defined by

$$s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor, \quad n \geq 0,$$

encoding the sequence as O when $s_n = 1$ and E when $s_n = 0$ (see also Section 4). Since $\alpha < 1$, we have $s_0 = \lfloor \alpha \rfloor = 0$, so the Sturmian word always begins with E.

DRAFT

12.1 Structural properties of mediant words

Before stating the main Sturmian identity we record four structural results about finite mediant words.

Theorem 12.1 (Every mediant word ends in O)**.** *For every estimate r produced by the mediant tree, the mediant word $\pi(r)$ ends in O.*

Proof. By structural induction on the mediant tree.

Base case. $\pi(e/1) = \mathsf{E}^e\mathsf{O}$, which ends in O.

Inductive step. Assume $\pi(r_0)$ and $\pi(r_1)$ both end in O. If $\Delta > 0$ then $\pi(r_{\text{new}}) = \pi(r_1)^w \cdot \pi(r_0)$, whose last symbol is the last symbol of $\pi(r_0)$, which is O by hypothesis. If $\Delta < 0$ then $\pi(r_{\text{new}}) = \pi(r_0) \cdot \pi(r_1)^w$, whose last symbol is the last symbol of the final copy of $\pi(r_1)$, again O. □

Theorem 12.2 (Overestimate criterion)**.** *Let $\theta > 1$ be irrational, let $r = (e, o)$ be a good estimate of θ (Definition 6.4), and set $L = e + o$. Let s be the Sturmian word with slope $\alpha = 1/(1+\theta)$.*

(1) *The Sturmian symbol s_{L-1} satisfies*

$$s_{L-1} = \begin{cases} \mathsf{O} & \text{if } e > \theta o \text{ (r an overestimate),} \\ \mathsf{E} & \text{if } e < \theta o \text{ (r an underestimate).} \end{cases}$$

(2) *If moreover r is produced by the mediant tree, then $\pi(r)[-1] = \mathsf{O}$ (Theorem 12.1), and therefore $\pi(r)[-1] = s_{L-1}$ if and only if r is an overestimate.*

Proof. **Part (1).** The Sturmian word satisfies $s_j = \mathsf{O}$ iff the fractional part $f_j := \{(j+1)\alpha\}$ lies in $[0, \alpha)$. So $s_{L-1} = \mathsf{O}$ iff $\{(e+o)\alpha\} \in [0, \alpha)$.

Set $K = o(e - \theta o)$, so $e = \theta o + K/o$. Since $\alpha = 1/(1+\theta)$:

$$(e+o)\alpha = o + \frac{K\alpha}{o}.$$

Since o is a positive integer, $\{(e+o)\alpha\} = \{K\alpha/o\}$. Since r is a good estimate, $|K|/o < 1$; with $\alpha < 1/2$ we have $|K\alpha/o| < \alpha < 1/2$, so:

- $K > 0$: $\{K\alpha/o\} = K\alpha/o \in (0, \alpha)$, so $s_{L-1} = \mathsf{O}$.
- $K < 0$: $\{K\alpha/o\} = 1 + K\alpha/o \in (1-\alpha, 1) \subset (\alpha, 1)$, so $s_{L-1} = \mathsf{E}$.

Part (2). For a tree estimate, $\pi(r)[-1] = \mathsf{O}$ by Theorem 12.1. Combined with Part (1), $\pi(r)[-1] = s_{L-1}$ iff both equal O, i.e. iff r is an overestimate. □

Remark 12.3. When $r = \text{Mediant}(r_{2d}, r_{2d+1}, w^+)$ is the child at the bracketing weight, the overestimate condition $e > \theta o$ is equivalent to $\Delta(r_{2d}, r_{2d+1}) > 0$: since w^+ is the bracketing weight, $\text{sgn}(e - \theta o) = \text{sgn}(f(w^+)) = \text{sgn}(B) = \text{sgn}(\Delta)$ (where $f(t) = A + tB$ as in the proof of Theorem 8.1, and $\Delta = \Delta(r_{2d}, r_{2d+1})$ is the determinant of the *parent pair*). For the companion child $r = \text{Mediant}(r_{2d}, r_{2d+1}, w^-)$ (when $w^+ = 1$, the companion is r_{2d} itself), the relationship is reversed. The sign of Δ of the parent pair is therefore not a uniform criterion for a general good estimate; the sign of $K = o(e - \theta o)$ is.

DRAFT

Theorem 12.4 (Universal EE prefix)**.** *For every non-leaf estimate r produced by the mediant tree (i.e. $|\pi(r)| > \lceil\theta\rceil + 1$), the mediant word begins $\pi(r)[0{:}2] = \mathsf{EE}$.*

Proof. By Lemma 9.3, $\pi(r) = \text{Ch}^-(e, o)$. Every non-leaf estimate $r = e/o$ lies strictly between the two leaf estimates $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$, so in particular $e/o > \lfloor\theta\rfloor \geq 1$, giving $e > o$. By Definition 5.1, $\text{Ch}^-(e, o)[i] = \mathsf{O}$ iff $\lfloor(i+1)o/(e+o)\rfloor > \lfloor i \cdot o/(e+o)\rfloor$. For $i = 0$: $1 \cdot o/(e+o) < 1$ (since $o < e + o$), so $\lfloor o/(e+o)\rfloor = 0 = \lfloor 0\rfloor$, giving $\pi(r)[0] = \mathsf{E}$. For $i = 1$: $e > o$ implies $e + o > 2o$, so $2o/(e+o) < 1$, giving $\lfloor 2o/(e+o)\rfloor = 0$, and $\pi(r)[1] = \mathsf{E}$. Hence $\pi(r)[0{:}2] = \mathsf{EE}$. □

Theorem 12.5 (Palindrome invariant)**.** *For every estimate r produced by the mediant tree with $L = |\pi(r)| \geq 2$, the interior slice $\pi(r)[1{:}L{-}1]$ is a palindrome.*

Proof. By Lemma 9.3, $\pi(r) = \text{Ch}^-(e, o)$ with $L = e{+}o$. By Definition 5.1, $\text{Ch}^-(e, o) = \mathsf{E}{\cdot}U{\cdot}\mathsf{O}$ where U is the palindromic interior (Berstel [5], Chapter 1). Therefore $\pi(r)[1{:}L{-}1] = U$ is a palindrome. □

Theorem 12.6 (Sturmian identity)**.** *Let $\theta > 1$ be irrational, $\alpha = 1/(\theta+1)$, and let s be the Sturmian sequence with slope α. For every estimate r produced by the mediant tree, with mediant word $\pi(r)$ of length N,*

$$\pi(r)[i] = s_i \qquad \text{for all } 0 \le i \le N-2.$$

The terminal position $i = N-1$ is excluded: by Theorem 12.1 the mediant always emits O *there, while by Theorem 12.2 the Sturmian emits* O *if r is an overestimate and* E *if r is an underestimate.*

Proof. By Lemma 9.3, $\pi(r) = \mathrm{Ch}^-(e,o)$ with $L = e+o$. By Definition 5.1, $\mathrm{Ch}^-(e,o)[i] = \mathsf{O}$ iff $\lfloor (i+1)o/L \rfloor > \lfloor i \cdot o/L \rfloor$, and the Sturmian word satisfies $s_i = \mathsf{O}$ iff $\lfloor (i+1)\alpha \rfloor > \lfloor i\alpha \rfloor$. It suffices to show $\lfloor n \cdot o/L \rfloor = \lfloor n\alpha \rfloor$ for all $1 \le n \le L-1$.

The estimate $r = e/o$ belongs to a bounded pair (r_{2d}, r_{2d+1}) at some depth d, with the two fractions $\alpha_0 = o_0/L_0$ and $\alpha_1 = o_1/L_1$ (where $L_k = e_k + o_k$) satisfying $\alpha_0 < \alpha < \alpha_1$ and the Farey adjacency condition $o_1 L_0 - o_0 L_1 = \pm 1$. In particular o/L is one of α_0 or α_1, and $L = L_k$ for the appropriate k.

Key lemma (Farey gap). For $1 \le n \le L-1$, no fraction with denominator $< L$ lies strictly between α and o/L. *Proof:* Suppose m/n with $0 < n < L$ lies strictly between α and o/L. Then m/n lies strictly between α_0 and α_1. By the Farey mediant property, any fraction strictly between two Farey-adjacent fractions $\alpha_0 = o_0/L_0$ and $\alpha_1 = o_1/L_1$ (with $|o_1 L_0 - o_0 L_1| = 1$) has denominator at least $L_0 + L_1 > L$, contradicting $n < L$.

Agreement. Since no fraction with denominator $< L$ lies strictly between α and o/L, for each $1 \le n \le L-1$ the two values $n\alpha$ and $n \cdot o/L$ lie in the same unit interval $[m, m+1)$ for some integer m, giving $\lfloor n\alpha \rfloor = \lfloor n \cdot o/L \rfloor$. Hence $\mathrm{Ch}^-(e,o)[i] = s_i$ for $0 \le i \le L-2$.

DRAFT

Terminal position. The exclusion of $i = L-1$ follows from Theorems 12.1 and 12.2: $\pi(r)$ ends in O, while $s_{L-1} = \mathsf{O}$ iff r is an overestimate ($e/o > \theta$). □

Corollary 12.7 (Mismatch characterisation)**.** *The mediant word $\pi(r)$ and the Sturmian prefix $s[0{:}L]$ are identical if and only if r is an overestimate ($e/o > \theta$). When r is an underestimate ($e/o < \theta$), they agree at every position except $i = L-1$, where the mediant emits* O *and the Sturmian emits* E*.*

The single terminal discrepancy is an artefact of truncation, not a genuine disagreement: the mediant sequence closes its last primitive word with O at position $N-1$, whereas the Sturmian word — which continues indefinitely — may still be in the interior of a longer word at that depth. As the depth increases and $N \to \infty$, both sequences converge to the same infinite word.

Remark 12.8. The identification of finite approximants of a Sturmian word with lower Christoffel words is classical (Lothaire [3], Berstel et al. [5]). The novel content of Theorem 12.6 is the proof via the mediant tree, which gives an explicit construction of these approximants together with a precise characterisation of the terminal discrepancy (the last-letter criterion, Theorems 12.1 and 12.2).

This has been verified computationally for several irrational values of θ, including $\sqrt{2}$, $\log_2 3$, $\log_2 5$, ϕ, and π. For $\theta = \log_2 3$ the verification extends to sequences of length up to 27,869,189 symbols: at every depth the two sequences agree everywhere except possibly the last symbol, where the mediant emits O and the Sturmian emits E.

Remark 12.9 (The mediant sequence is the better finite approximation). At depths where the two sequences differ, the mediant's finite mediant word is a closer approximation to $\alpha = 1/(\theta + 1)$ in terms of running O-density. The terminal O emitted by the mediant sequence gives a density slightly *above* α, whereas the Sturmian's terminal E gives a density substantially *below* α.

This reflects a structural asymmetry in the two methods. The Sturmian algorithm at position $N - 1$ is mid-word in a longer primitive word, so it emits E — inherently favouring the above-θ estimate and leaving the running density below α. The mediant tree, by contrast, explicitly selects at each step the estimate with smaller $|\varepsilon|$, i.e. r_{2d+1}, whichever side of θ it falls on. Closing the sequence with O at that point reflects this selection: it is the correct terminal for a word whose ratio e/o slightly underestimates θ, nudging the running density up toward α.

For $\theta = \log_2 3$, the improvement at the discrepancy depths is substantial: at $N = 1719$ the mediant density error is 1.4×10^{-8} against the Sturmian's 5.8×10^{-4} — roughly four orders of magnitude better.

The O-density of $\pi(r)$ for an estimate e/o is

$$\frac{o}{e+o} = \frac{1}{e/o+1} \xrightarrow{e/o\to\theta} \frac{1}{\theta+1} = \alpha,$$

consistent with the Sturmian slope.

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Efficiency and lazy generation. Although the mediant words are Sturmian, the mediant tree is a far more efficient way to generate them. To produce a prefix of N symbols, the naive Sturmian algorithm must evaluate $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ at each of the N positions, requiring $O(N)$ real-arithmetic operations.

The mediant algorithm instead builds a compressed representation of the same prefix. Each iteration requires a constant number of real-arithmetic operations to compute the next bounded pair and weight; after k iterations the tree encodes a Sturmian prefix of length $N = e_{2d+1} + o_{2d+1}$. Empirically N grows rapidly with k (see Section 10), suggesting $O(\log N)$ real-arithmetic operations suffice to encode a prefix of length N, though a formal proof of this bound remains open.

Once the tree is built, enumerating the N symbols requires *no real arithmetic at all*: each node specifies an integer weight w and two child words, and reading off the sequence is a recursive depth-first traversal governed entirely by integer counts. No auxiliary memory beyond the call stack is required.

OO-avoidance. Since $\theta > 1$ implies $\alpha = 1/(\theta + 1) < 1/2$, the Sturmian word avoids OO: two consecutive O's would require $\lfloor (n+2)\alpha \rfloor - \lfloor n\alpha \rfloor = 2$, but $2\alpha < 1$ makes this impossible. The mediant tree gives an independent structural proof: every mediant word ends in O and begins with E (since $e \geq 1$; Lemma 9.3 and Definition 5.1), so every concatenation boundary

within a branch word $v^w \cdot u$ or $u \cdot v^w$ reads $\cdots \mathsf{OE} \cdots$; OO never arises at any boundary, and by induction never arises anywhere.

Remark 12.10. The weights w_{2d+1} are related to the partial quotients of the continued fraction expansion of θ, though they are not in general identical to them. A large weight at step n corresponds to a large partial quotient, producing a long run of one symbol type in the mediant word — the same phenomenon that gives rise to the irregular spacing of O's and E's in an irrational Sturmian word. Conversely, small weights correspond to small partial quotients, and the mediant word advances by only a short block at each such step.

13 Extension to $0 < \theta < 1$

The algorithm of Sections 7–9 requires $\theta > 1$. We now show how to handle $\theta \in (0, 1)$ by algebraic reduction, in two stages.

If one were content to accept OO in the mediant word, no special treatment would be needed: the algorithm and the Sturmian definition extend formally to any $\theta > 0$, and for $\theta < 1$ the slope $\alpha = 1/(1+\theta) > 1/2$ makes OO unavoidable. The reductions below are motivated by applications (described in [6]) that require OO-free mediant words. More concretely: for $\theta \in (0, 1)$, setting $\theta' = (1-\theta)/\theta$ (when $\theta < 1/2$) or $\theta' = \theta/(1-\theta)$ (when $\theta > 1/2$) maps θ to $\theta' > 1$, giving a slope $1/(1+\theta') < 1/2$ and hence an OO-free sequence; the counts e and o from that sequence then yield a rational approximant to the original θ.

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13.1 Case $0 < \theta < \frac{1}{2}$

For $\theta \in (0, \frac{1}{2})$, define

$$\theta' = \frac{1-\theta}{\theta} = \frac{1}{\theta} - 1 > 1.$$

Run the algorithm on θ'. A bounded pair $e/o \approx \theta'$ satisfies $e/o \approx (1-\theta)/\theta$, so

$$\theta \approx \frac{o}{e+o}.$$

The approximant to θ is therefore $o/(e+o)$, with numerator o (the O-count of the mediant word) and denominator $e+o$ (its total length).

Mediant word. The mediant word for θ' has O-density

$$\frac{1}{1+\theta'} = \frac{1}{1+(1-\theta)/\theta} = \theta,$$

which is less than $\frac{1}{2}$ since $\theta < \frac{1}{2}$. Consequently the OO-avoidance result of Section 12 applies: the mediant word for θ' is OO-free, and its O-density converges to θ.

Sturmian identity. The Sturmian word with slope $\alpha' = \theta$ is the infinite mediant word produced by the mediant tree for θ' (Theorem 12.6). This Sturmian word encodes θ directly: the O-density of any long prefix approximates θ with the same precision as the underlying mediant approximation of θ'.

13.2 Case $\frac{1}{2} < \theta < 1$

For $\theta \in (\frac{1}{2}, 1)$, note that $1 - \theta \in (0, \frac{1}{2})$, so define

$$\theta' = \frac{\theta}{1-\theta} > 1$$

and run the algorithm on θ'. A bounded pair $e/o \approx \theta'$ satisfies $e/o \approx \theta/(1-\theta)$, so

$$\theta \approx \frac{e}{e+o}.$$

No modification of the mediant word is required: the sequence for θ' is used solely to produce the counts e and o, which then yield the approximant $e/(e+o)$ to the original θ.

Mediant word. The mediant word for θ' has O-density

$$\frac{1}{1+\theta'} = \frac{1}{1+\theta/(1-\theta)} = 1 - \theta < \tfrac{1}{2},$$

so it is OO-free and its Sturmian slope is $1 - \theta$. The sequence encodes θ'; the rational approximant $e/(e+o)$ to θ is read off from the E- and O-counts of that sequence.

Summary. Table 1 collects the reductions for all ranges of θ.

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Table 1: Reductions to the $\theta > 1$ algorithm for all ranges of θ (four non-trivial, three trivial boundary cases)

Range of θ	Reduction θ'	Approximant to θ	O-density
$\theta > 1$	$\theta' = \theta$	e/o	$1/(1+\theta) < \frac{1}{2}$
$0 < \theta < \frac{1}{2}$	$\theta' = (1-\theta)/\theta$	$o/(e+o)$	$\theta < \frac{1}{2}$
$\frac{1}{2} < \theta < 1$	$\theta' = \theta/(1-\theta)$	$e/(e+o)$	$1 - \theta < \frac{1}{2}$
$\theta < 0$	$\theta' = -\theta$ (or above)	negate	—
$\theta = 0$	trivial: word E, $o/(o+e) = 0$	$0/1$	—
$\theta = \frac{1}{2}$	trivial: word EO, $o/(o+e) = \frac{1}{2}$	$1/2$	—
$\theta = 1$	trivial: word O, $o/(o+e) = 1$	$1/1$	—

14 Conclusion and Open Problems

We have presented a theory of rational approximation for real numbers based on bounded pairs and the weighted mediant construction. The core algorithm applies to $\theta > 1$; all other real numbers are handled by algebraic reduction (Section 13).

What is new. The weighted mediant step is the classical Stern–Brocot step; the novelty is Theorem 8.1, which identifies the *unique optimal weight* w^+ via a sign-change bracket on the linear error function, giving a closed-form formula and guaranteeing that each iterate is at least a Stern–Brocot semi-convergent — the best approximant achievable in one step from the current bounded pair. Lemma 9.3 (every mediant word is a lower Christoffel word) follows from the standard Christoffel concatenation theorem (Berstel et al. [5]); the contribution is the identification that the mediant algorithm *naturally* produces lower Christoffel words, without any explicit reference to the Christoffel tree. The Sturmian identity (Theorem 12.6) and the palindrome invariant (Theorem 12.5) are consequences of the Christoffel word theory (Lothaire [3], Berstel et al. [5]); the contribution is a direct proof via the mediant construction, together with the last-letter criterion (Theorems 12.1 and 12.2) which is new. The mediant tree taken as a whole is a *compressed transducer*: it encodes the construction history of a length-N Sturmian prefix in $O(\log N)$ integer-arithmetic steps (empirically; a formal bound remains open), after which the symbols are enumerated by a purely integer recursive traversal.

Open problems.

1. **Superlinear denominator growth.** Prove that o_{2d+1} grows superlinearly (or exponentially) with the iteration count. The empirical evidence is strong — every tested irrational shows at least $4\times$ growth per two steps — but a formal proof is lacking.

2. **Characterisation of the MT fractions.** Every MT fraction is at least a Stern–Brocot semi-convergent; the MT visits the subsequence of those whose consecutive mediants straddle θ. Characterise this subsequence in terms of the continued-fraction expansion of θ, and determine when an MT fraction equals a CF convergent.

3. **Cost comparison with CF.** The CF algorithm produces convergents in $O(\log N)$ arithmetic steps; the MT does likewise but visits different fractions. Give a precise comparison of the approximation quality per step between the two algorithms.

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A Proof of Theorem 8.1

We prove the closed-form formula $w^+ = \lfloor -A/B \rfloor + 1$ directly from the bracketing condition, using only a linear sign argument.

We have a bounded pair (r_{2d}, r_{2d+1}) bracketing θ, with

$$A = e_{2d} - \theta\, o_{2d}, \qquad B = e_{2d+1} - \theta\, o_{2d+1}.$$

Since r_{2d} and r_{2d+1} are on opposite sides of θ we have $\operatorname{sgn}(A) = -\operatorname{sgn}(B)$, so $A/B < 0$ and $c := -A/B > 0$. The two weighted mediants are

$$r_- = \frac{e_{2d} + (w-1)\, e_{2d+1}}{o_{2d} + (w-1)\, o_{2d+1}}, \qquad r_+ = \frac{e_{2d} + w\, e_{2d+1}}{o_{2d} + w\, o_{2d+1}},$$

with signed errors (numerators of ε_-, ε_+ up to positive denominators)

$$A + (w-1)B \quad \text{and} \quad A + wB.$$

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Bracketing condition. The pair (r_-, r_+) brackets θ iff ε_- and ε_+ have opposite signs, i.e. $(A+(w-1)B)(A+wB) < 0$. Writing $c = -A/B$, this product equals $B^2\,(w-1-c)(w-c)$, which is negative iff $w-1-c$ and $w-c$ have opposite signs, i.e.

$$w - 1 < c < w \qquad \Longleftrightarrow \qquad c - 1 < w - 1 \le c \qquad \Longleftrightarrow \qquad w = \lfloor c \rfloor + 1.$$

Since θ is irrational, $A + nB \neq 0$ for all integers n, so c is never an integer and the condition is strict: exactly one integer w satisfies it, namely

$$w^+ = \lfloor -A/B \rfloor + 1.$$

We also write $w^- = w^+ - 1 = \lfloor -A/B \rfloor$ for the companion weight.

Expression in terms of Δ and K. Recall $K = o_{2d+1}B$ and $A\, o_{2d+1} - B\, o_{2d} = -\Delta$, giving $A = (Bo_{2d} - \Delta)/o_{2d+1}$, so

$$\frac{-A}{B} = \frac{\Delta}{o_{2d+1}B} - \frac{o_{2d}}{o_{2d+1}} = \frac{\Delta}{K} - \frac{o_{2d}}{o_{2d+1}},$$

and therefore

$$w^+ = \left\lfloor \frac{\Delta}{K} - \frac{o_{2d}}{o_{2d+1}} \right\rfloor + 1,$$

completing the proof of Theorem 8.1.

Termination. When $K = 0$, r_{2d+1} is an exact rational representation of θ; no further improvement is possible and the algorithm terminates.

B Examples and Computational Results

B.1 Worked example: mediant words for $\log_2 3$

Example B.1 (Mediant words and trees for $\log_2 3$)**.** Starting from the initial bounded pair $(r_0, r_1) = (1/1,\ 2/1)$ with $\theta = \log_2 3 \approx 1.58496$, the algorithm produces the following sequence of bounded pairs. At each depth d, weight w_{2d+1} is used to compute the two candidates $r_- = \text{Mediant}(r_{2d}, r_{2d+1}, w_{2d+1} - 1)$ and $r_+ = \text{Mediant}(r_{2d}, r_{2d+1}, w_{2d+1})$; the better estimate (smaller $|\varepsilon|$) becomes r_{2d+3}.

d	r_{2d}	r_{2d+1}	$\varepsilon_{r_{2d}}$	$\varepsilon_{r_{2d+1}}$	w_{2d+1}	r_-	r_+
0	1/1	2/1	-0.369	$+0.262$	2	3/2	5/3
1	3/2	5/3	-0.054	$+0.052$	1	3/2	8/5
2	3/2	8/5	-0.054	$+0.009$	3	19/12	27/17
3	27/17	19/12	$+0.002$	-0.001	3	65/41	84/53

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The leaf words are $\pi(r_0) =$ `EO` and $\pi(r_1) =$ `EEO`. Reading the word decomposition bottom-up:

Level 1 ($d = 0$, $w_1 = 2$, $\Delta > 0$, $\pi(r_3) = \pi(r_1)^2 \cdot \pi(r_0)$, $\pi(r_2) = \pi(r_1)^1 \cdot \pi(r_0)$):

$$\underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\ \underbrace{\texttt{EEO}}_{\pi(r_1)}\ \underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_3)=\texttt{EEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\ \underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_2)=\texttt{EEOEO}}$$

Level 2 ($d = 1$, $w_3 = 1$, $\Delta > 0$, $\pi(r_5) = \pi(r_3)^1 \cdot \pi(r_2)$, $\pi(r_4) = \pi(r_2)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEO}}_{\pi(r_3)}\ \underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_5)=\texttt{EEOEEOEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_4)=\texttt{EEOEO}}$$

Level 3 ($d = 2$, $w_5 = 3$, $\Delta > 0$, $\pi(r_7) = \pi(r_5)^2 \cdot \pi(r_4)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\ \underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\ \underbrace{\texttt{EEOEO}}_{\pi(r_4)}}_{\pi(r_7)}$$

The same estimates as mediant trees (depths $d = 1$ and $d = 2$):

$$r_2 = 3/2 = \text{Mediant}(\underbrace{1}_{\text{leaf}},\ \underbrace{2}_{\text{leaf}},\ w = 1),$$

$$r_3 = 5/3 = \text{Mediant}(\underbrace{1}_{\text{leaf}},\ \underbrace{2}_{\text{leaf}},\ w = 2),$$

$$r_4 = 3/2 = \text{Mediant}(\text{Mediant}(1,\ 2,\ 1),\ \text{Mediant}(1,\ 2,\ 2),\ w = 0),$$

$$r_5 = 8/5 = \text{Mediant}(\text{Mediant}(1,\ 2,\ 1),\ \text{Mediant}(1,\ 2,\ 2),\ w = 1).$$

A weight of $w = 0$ means the left subtree is used directly (contributing zero copies of the right subtree).

B.2 Approximating $\log_2 3$

Table 2 shows 22 steps of the algorithm for $\theta = \log_2 3$, starting from $r_0 = 1/1$, $r_1 = 2/1$. Note the large weight $w = 24$ at depth $d = 6$ and $w = 55$ at depth $d = 10$, corresponding to large partial quotients of the continued fraction of $\log_2 3$, which produce large jumps in precision.

Table 2: Successive bounded pairs approximating $\log_2 3 \approx 1.58496\ldots$

d	r_{2d}	r_{2d+1}	$\varepsilon_{r_{2d}}$	$\varepsilon_{r_{2d+1}}$	w^+	r_-	r_+
0	1/1	2/1	-0.369	$+0.262$	2	3/2	5/3
1	3/2	5/3	-0.054	$+0.052$	1	3/2	8/5
2	3/2	8/5	-0.054	$+0.009$	3	19/12	27/17
3	27/17	19/12	$+0.002$	-0.001	3	65/41	84/53
4	65/41	84/53	$+2.54 \times 10^{-4}$	-3.59×10^{-5}	6	485/306	569/359
5	485/306	569/359	$+3.04 \times 10^{-6}$	-2.70×10^{-6}	1	485/306	1054/665
6	485/306	1054/665	$+3.04 \times 10^{-6}$	-5.98×10^{-8}	24	24727/15601	25781/16266
7	25781/16266	24727/15601	-1.42×10^{-9}	$+1.06 \times 10^{-9}$	2	50508/31867	75235/47468
8	75235/47468	50508/31867	$+2.10 \times 10^{-10}$	-2.08×10^{-10}	2	125743/79335	176251/111202
9	125743/79335	176251/111202	$+4.20 \times 10^{-11}$	-2.95×10^{-11}	2	301994/190537	478245/301739
10	478245/301739	301994/190537	-1.07×10^{-11}	$+3.08 \times 10^{-13}$	55	16785921/10590737	17087915/10781274
11	16785921/10590737	17087915/10781274	-4.50×10^{-15}	$+1.03 \times 10^{-15}$	5	85137581/53715833	102225496/64497107
12	102225496/64497107	85137581/53715833	$+1.23 \times 10^{-16}$	-5.88×10^{-17}	3	272500658/171928773	357638239/225644606
13	272500658/171928773	357638239/225644606	$+9.47 \times 10^{-18}$	-6.79×10^{-18}	2	630138897/397573379	987777136/623217985
14	987777136/623217985	630138897/397573379	-2.30×10^{-18}	$+2.42 \times 10^{-19}$	15	9809721694/6189245291	10439860591/6586818670
15	9809721694/6189245291	10439860591/6586818670	-1.41×10^{-20}	$+1.40 \times 10^{-21}$	10	103768467013/65470613321	114208327604/72057431991
16	114208327604/72057431991	103768467013/65470613321	$+6.96 \times 10^{-23}$	-6.41×10^{-23}	2	217976794617/137528045312	321745261630/202998658633
17	321745261630/202998658633	217976794617/137528045312	-1.67×10^{-23}	$+5.95 \times 10^{-24}$	5	1193652440098/753110839881	1411629234715/890638885193
18	1411629234715/890638885193	1193652440098/753110839881	$+7.97 \times 10^{-25}$	-1.44×10^{-25}	7	8573543875303/5409303924479	9767196315401/6162414764360
19	8573543875303/5409303924479	9767196315401/6162414764360	$+1.11 \times 10^{-26}$	-7.83×10^{-27}	2	18340740190704/11571718688839	28107936506105/17734133453199
20	28107936506105/17734133453199	18340740190704/11571718688839	-2.06×10^{-27}	$+1.01 \times 10^{-27}$	4	83130157078217/52449289519716	101470897268921/64021008208555
21	101470897268921/64021008208555	83130157078217/52449289519716	$+1.63 \times 10^{-28}$	-2.51×10^{-29}	8	683381996816440/431166034846567	766512153894657/483615324366283

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B.3 Approximating π

Table 3 shows 22 steps for $\theta = \pi$, starting from $r_0 = 4/1$, $r_1 = 3/1$. The classical approximant $355/113$ appears at depth $d = 1$ (as r_+) with $\varepsilon \approx +8.49 \times 10^{-8}$, and the weight $w = 293$ at depth $d = 2$ produces an immediate jump of roughly three orders of magnitude (from $|\varepsilon| \approx 8.49 \times 10^{-8}$ to $|\varepsilon| \approx 1.06 \times 10^{-10}$).

Table 3: Successive bounded pairs approximating $\pi \approx 3.14159\ldots$

d	r_{2d}	r_{2d+1}	$\varepsilon_{r_{2d}}$	$\varepsilon_{r_{2d+1}}$	w^+	r_-	r_+
0	4/1	3/1	$+0.273$	-0.045	7	22/7	25/8
1	25/8	22/7	-5.0×10^{-3}	$+4.02 \times 10^{-4}$	15	333/106	355/113
2	333/106	355/113	-2.65×10^{-5}	$+8.49 \times 10^{-8}$	293	103993/33102	104348/33215
3	103993/33102	104348/33215	-1.84×10^{-10}	$+1.06 \times 10^{-10}$	2	208341/66317	312689/99532
4	208341/66317	312689/99532	-3.89×10^{-11}	$+9.28 \times 10^{-12}$	3	833719/265381	1146408/364913
5	833719/265381	1146408/364913	-2.77×10^{-12}	$+5.13 \times 10^{-13}$	4	4272943/1360120	5419351/1725033
6	4272943/1360120	5419351/1725033	-1.29×10^{-13}	$+7.05 \times 10^{-15}$	15	80143857/25510582	85563208/27235615
7	85563208/27235615	80143857/25510582	$+2.74 \times 10^{-16}$	-1.84×10^{-16}	2	165707065/52746197	245850922/78256779
8	165707065/52746197	245850922/78256779	$+5.22 \times 10^{-17}$	-2.49×10^{-17}	2	411557987/131002976	657408909/209259755
9	411557987/131002976	657408909/209259755	$+6.16 \times 10^{-18}$	-5.45×10^{-18}	1	411557987/131002976	1068966896/340262731
10	411557987/131002976	1068966896/340262731	$+6.16 \times 10^{-18}$	-9.77×10^{-19}	3	2549491779/811528438	3618458675/1151791169
11	2549491779/811528438	3618458675/1151791169	$+1.76 \times 10^{-19}$	-1.65×10^{-19}	1	2549491779/811528438	6167950454/1963319607
12	2549491779/811528438	6167950454/1963319607	$+1.76 \times 10^{-19}$	-2.43×10^{-20}	3	14885392687/4738167652	21053343141/6701487259
13	14885392687/4738167652	21053343141/6701487259	$+9.94 \times 10^{-21}$	-8.33×10^{-23}	85	1783366216531/567663097408	1804419559672/574364584667
14	1804419559672/574364584667	1783366216531/567663097408	-5.85×10^{-25}	$+3.91 \times 10^{-25}$	2	3587785776203/1142027682075	5371151992734/1709690779483
15	3587785776203/1142027682075	5371151992734/1709690779483	-1.00×10^{-25}	$+6.28 \times 10^{-26}$	2	8958937768937/2851718461558	14330089761671/4561409241041
16	14330089761671/4561409241041	8958937768937/2851718461558	$+2.20 \times 10^{-26}$	-2.46×10^{-27}	15	139755218526789/44485467702853	148714156295726/47337186164411
17	148714156295726/47337186164411	139755218526789/44485467702853	-9.99×10^{-29}	$+5.13 \times 10^{-29}$	3	428224593349304/136308121570117	567979811876093/180793589272970
18	567979811876093/180793589272970	428224593349304/136308121570117	$+1.17 \times 10^{-29}$	-1.21×10^{-30}	13	5706674932067741/1816491048114374	6134899525417045/1952799169684491
19	5706674932067741/1816491048114374	6134899525417045/1952799169684491	$+7.43 \times 10^{-32}$	-1.55×10^{-32}	5	30246273033735921/9627687726852338	36381172559152966/11580486896536829
20	30246273033735921/9627687726852338	36381172559152966/11580486896536829	$+1.45 \times 10^{-33}$	-1.40×10^{-33}	1	30246273033735921/9627687726852338	66627445592888887/21208174623389167
21	30246273033735921/9627687726852338	66627445592888887/21208174623389167	$+1.45 \times 10^{-33}$	-1.07×10^{-34}	7	430010946591069243/136876735467187340	4966383921839581 30/158084910090576507